assessment—not supported by any intelligence—that *Tirpitz*, *Hipper*, and *Scheer* were moving to intercept the convoy. The First Sea Lord Sir Dudley Pound ordered the convoy to scatter, concentrating all the escort and covering warships to meet the German sortie. It was an appalling error. The German warships never appeared, having returned to harbor after the briefest of forays. Meanwhile, the scattered convoy, escorted only by a handful of corvettes, minesweepers, and armed trawlers, was defenseless against the remorseless attacks of the Luftwaffe and the U-boats.

SITTING TARGETS

On July 5 12 merchant ships were sunk, seven of them American. Six of the sinkings were claimed by U-boats and six by the Luftwaffe. Over the next two days five more ships went to the bottom, three of them victims of a single U-boat, Captain Reinhardt Reche's U-255. A final two sinkings occurred on July 10, after which the remnants of the convoy crawled into Arkhangelsk or Murmansk. Out of the original 35 vessels, 22 had been sunk, among the worst losses of any convoy in the war. Along with 430 tanks, 200 aircraft, and almost 100,000 tons of cargo, the lives of 153 seamen had been lost. Many survivors endured days in open lifeboats in Arctic conditions before being rescued.

Freezing conditions
The icebound deck of a British cruiser sailing the convoy route between Britain and Russia in 1943 gives an idea of the risks and hardships facing sailors on the Arctic convoys.

WORLD WAR II

CONVOY HX-229 AND CONVOY SC-122

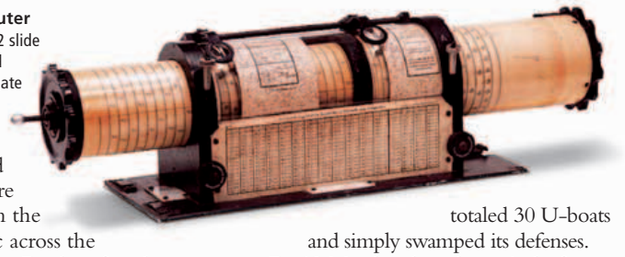


Date March 16–19, 1943
Forces Germans: 38 U-boats; Allies: 17 escort ships, 100 merchant vessels
Losses Germans: 1 U-boat; Allies: 22 merchant ships

Location North Atlantic Ocean

In early March 1943 two convoys left New York bound for Liverpool: HX-229 with 40 merchant ships and the 60 slower vessels of SC-122. Admiral Karl Dönitz knew of these convoys because B-Dienst intelligence department had cracked British naval codes. Three U-boat wolf packs, codenamed Raubgraf, Stürmer, and Dränger, were positioned in the mid-Atlantic across the convoys' route. Raubgraf made contact with HX-229 on March 16. The merchant ships had a weak escort of three destroyers and two corvettes, insufficient to prevent the eight-boat wolf pack from harrying the convoy mercilessly. Meanwhile, on their way to join the attack on HX-229, the other wolf packs stumbled upon SC-122. This had a stronger escort, but the wolf packs totaled 30 U-boats and simply swamped its defenses. By 19 March, when the arrival of more escorts halted the attacks, HX-229 had lost 13 ships and SC-122 had lost nine. Some 300 merchant seamen were killed. The Germans lost a single U-boat to an air patrol. A Royal Navy report later stated that "there seemed real danger the enemy would achieve his aim of severing the route which united Great Britain with the North American continent."

U-boat computer
This German HR2 slide computer helped U-boats to navigate using the stars.



CREW PROFILE

U-BOAT CREW

WORLD WAR II

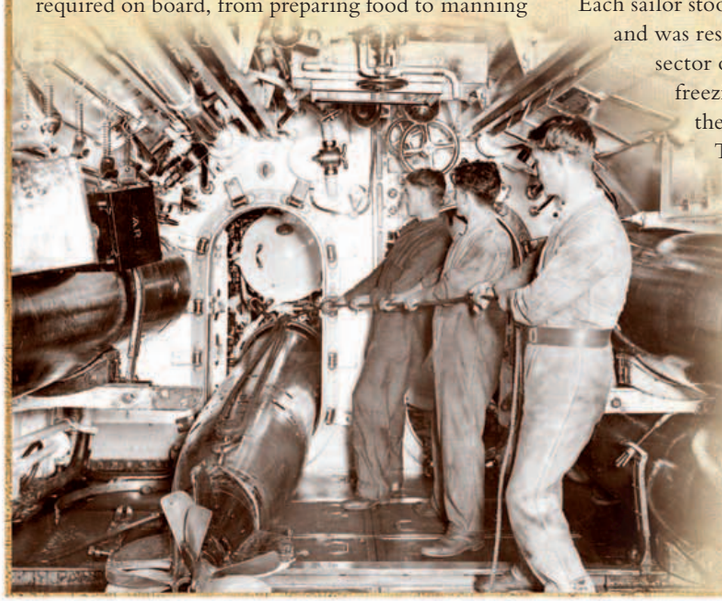
A TYPE VII, the standard U-boat of World War II, typically had a crew of four officers—the captain, his two lieutenants or "watch officers", and an engineering officer—and some 40 petty officers and seamen. The petty officers were responsible for various specialized areas such as crew discipline, the torpedoes, the diesel and electric engines, the radio, navigation, and steering. At least in the early period of the war, U-boat crews were a self-conscious elite, with five years' intensive training before going operational. They needed to be exceptionally fit and well balanced to endure the rigors of a submariner's life. Most of the men were in their early 20s, and their captains were usually not much older.

U-BOAT SEAMAN

The seamen were housed in the bows of the boat, a space they shared with the forward torpedoes. There was usually one bunk for every two men, occupied on a shift system as one was on duty while the other rested. There was no privacy and little opportunity to wash or get a change of clothes while on patrol. The seamen performed the whole range of tasks required on board, from preparing food to manning the guns and loading the torpedoes in their tubes. Their most onerous task was keeping watch. When the U-boat was on the surface there had to be four seamen on watch 24 hours a day, searching the horizon for a convoy to attack or a hostile warship or aircraft.

UNITY IN ADVERSITY

Each sailor stood at each corner of the conning tower and was responsible for observing a 90-degree sector of the sea. A four-hour watch, often in freezing cold and with waves sweeping over the bridge, could be a feat of endurance. To avoid being swept overboard in rough weather seamen on watch wore safety belts that tied them to the column in the middle of the conning tower, as well as special waterproof suits. In the claustrophobic, cramped conditions on a U-boat, most crews became like a family for the seamen – one U-boat man wrote that "you felt engulfed by comradeship."



Torpedo ratings
Crew members load a torpedo into one of the tubes on board a German U-boat. Seamen usually had to share their sleeping quarters with spare torpedoes.